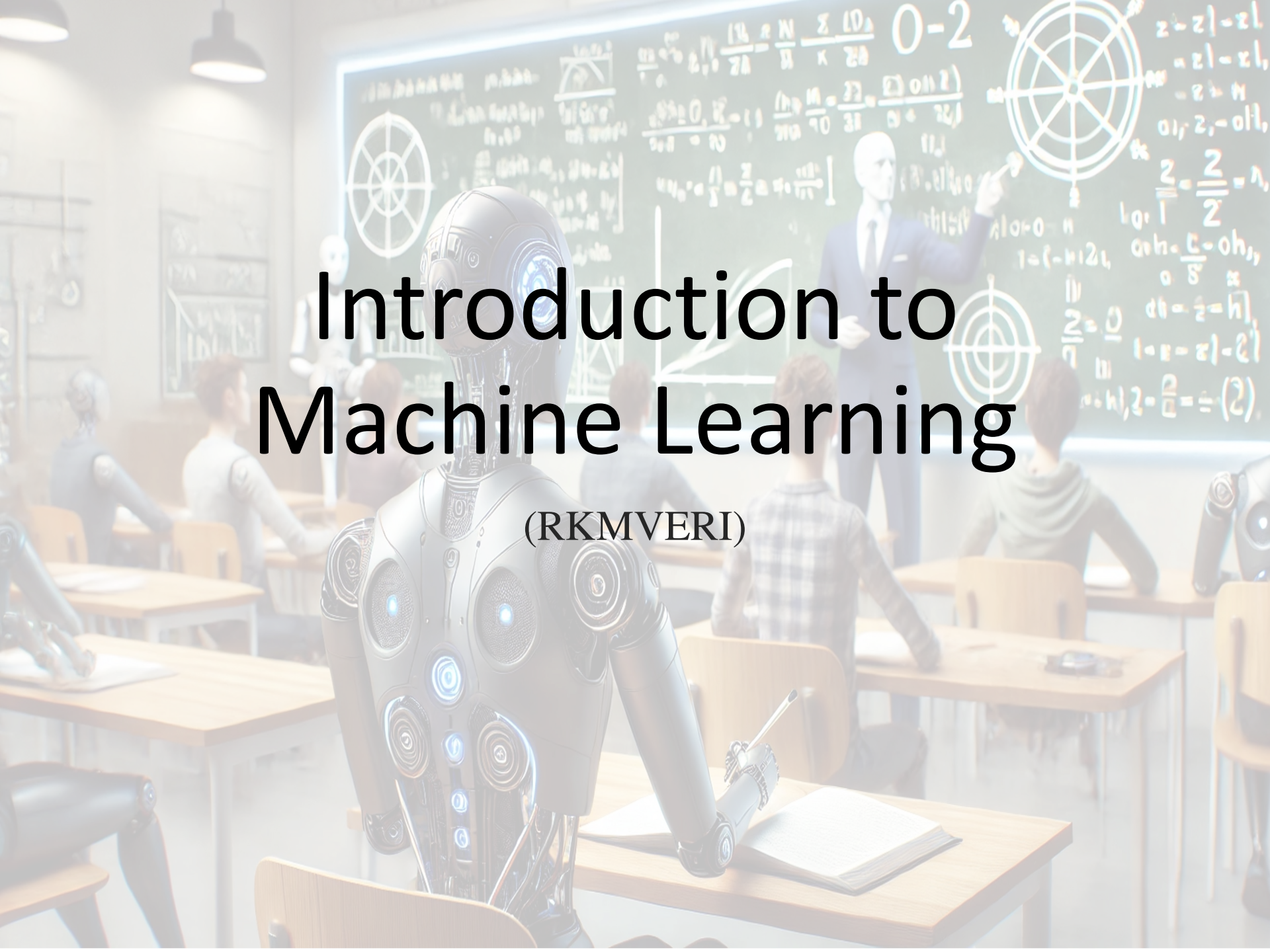




The background image is a conceptual illustration of a classroom. In the foreground, a robot with a metallic, segmented body and glowing blue circular lights on its chest and joints is seated at a wooden desk, holding a pen over an open book. In the background, a teacher robot in a blue suit stands at a chalkboard, pointing at it. The chalkboard is filled with various mathematical formulas, including the quadratic formula
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
, the area of a circle
$$A = \pi r^2$$
, and the volume of a sphere
$$V = \frac{4}{3}\pi r^3$$
. There are also diagrams of a target and a clock face. The overall scene is dimly lit, with a warm, yellowish glow from the desk lamps.

Introduction to Machine Learning

(RKMVERI)

Shanti Mantra

ॐ सह नाववतु ।

सह नौ भुनक्तु ।

सह वीर्यं करवावहै ।

तेजस्वि नावधीतमस्तु मा विद्विषावहै ।

ॐ शान्तिः शान्तिः शान्तिः ॥

Om Saha Nau-Avatu |

Saha Nau Bhunaktu |

Saha Viiryam Karavaavahai |

Tejasvi Nau-Adhiitam-Astu Maa Vidvissaavahai |

Om Shaantih Shaantih Shaantih ||

Om, may He protect us both (the teacher and the student).

May He nourish us both.

May we work together with energy and vigor.

May our study be enlightening and not give rise to hostility.

Om, peace, peace, peace.

Recap: Linear Regression

Gradient Descent varieties

- **Batch Gradient Descent:** Uses the entire dataset to compute the gradient at each step, ensuring stable convergence but can be slow for large datasets.
- **Stochastic Gradient Descent:** Updates parameters using a single data point at each step, allowing faster updates but with more variance and potential oscillations.
- **Mini-Batch Gradient Descent:** Combines the benefits of both by updating parameters using a small random subset of the data, balancing speed and stability.

SVD (Singular Value Decomposition) way

Normal Equation

$$\theta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Directly computing the inverse of $\mathbf{X}^T \mathbf{X}$ can be numerically unstable, especially if $\mathbf{X}^T \mathbf{X}$ is nearly singular or poorly conditioned. This is where SVD comes in.

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

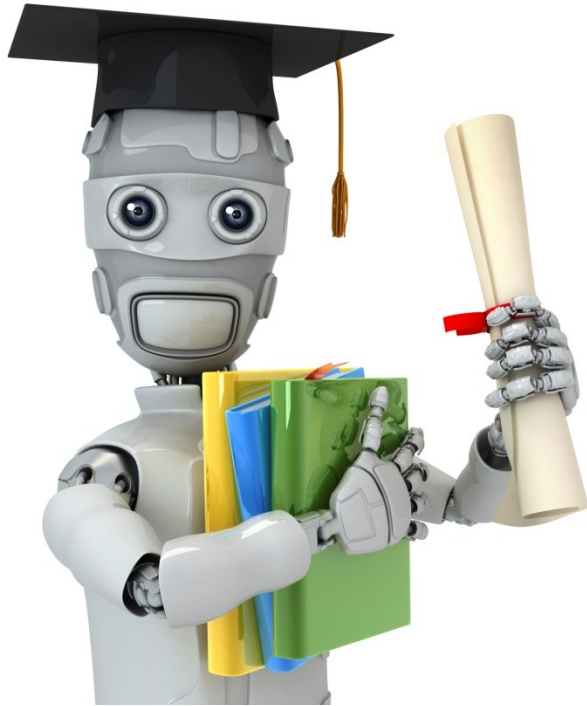
Where:

- $\mathbf{U}$ is an $m \times m$ orthogonal matrix (columns are left singular vectors of $\mathbf{X}$).
- $\mathbf{\Sigma}$ is an $m \times n$ diagonal matrix with non-negative real numbers on the diagonal (the singular values of $\mathbf{X}$).
- $\mathbf{V}^T$ is the transpose of an $n \times n$ orthogonal matrix (rows are right singular vectors of $\mathbf{X}$).

In SVD way:
$$\theta = \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^T \mathbf{y}$$

Comparison of algorithms

Algorithm	Large m (Instance)	Out- of-core support	Large n (Features)	Hyper- params	Scaling	Scikit-Learn
Normal equation	Fast	No	Slow	0	No	N/A
SVD	Fast	No	Slow	0	No	LinearRegression
Batch GD	Slow	No	Fast	2	Yes	N/A
Stochastic GD	Fast	Yes	Fast	≥ 2	Yes	SGDRegressor
Mini-batch GD	Fast	Yes	Fast	≥ 2	Yes	N/A



Machine Learning

Logistic Regression

Classification

Logistic Regression

- Some regression algorithms can be used for classification (and vice versa).
- Logistic regression (also called logit regression) is commonly used to estimate the probability that an instance belongs to a particular class
 - e.g., what is the probability that this email is spam?
- If the estimated probability is greater than a given threshold (typically 50%), then the model predicts that the instance belongs to that class (called the positive class, labeled “1”), and
- otherwise it predicts that it does not (i.e., it belongs to the negative class, labeled “0”).
- This makes it a binary classifier.

Classification

Email: Spam / Not Spam?

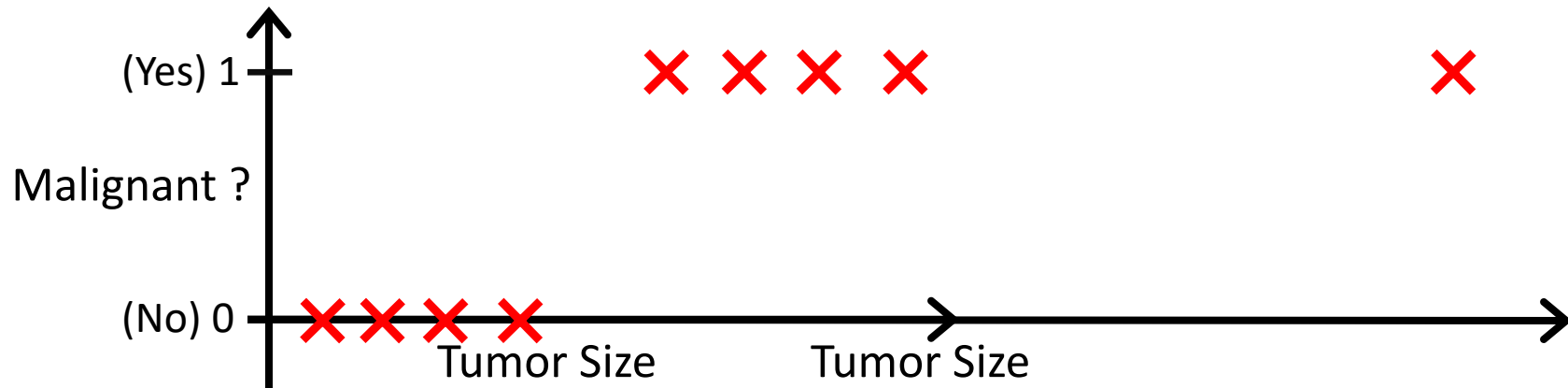
Online Transactions: Fraudulent (Yes / No)?

Tumor: Malignant / Benign ?

$$y \in \{0, 1\}$$

0: “Negative Class” (e.g., benign tumor)

1: “Positive Class” (e.g., malignant tumor)



Threshold classifier output $h_{\theta}(x)$ at 0.5:

If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

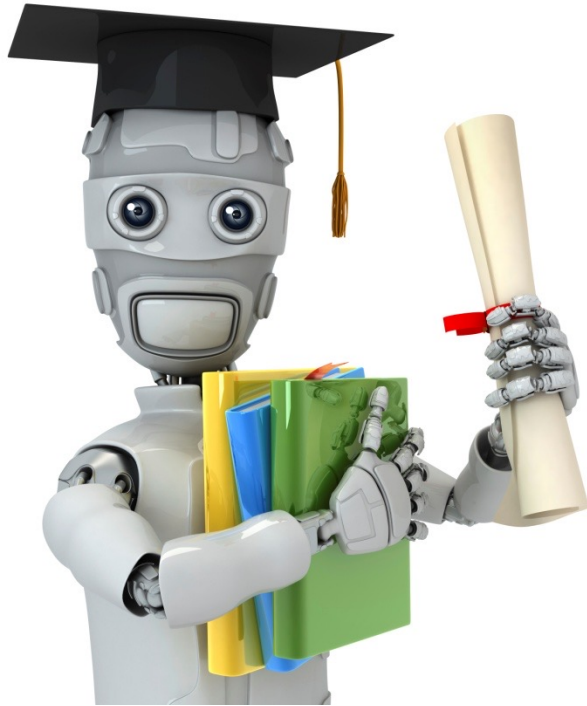
If $h_{\theta}(x) < 0.5$, predict "y = 0"

Classification: $y = 0$ or 1

$h_{\theta}(x)$ can be > 1 or < 0

Linear Regression

Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$



Machine Learning

Logistic
Regression

Hypothesis
Representation

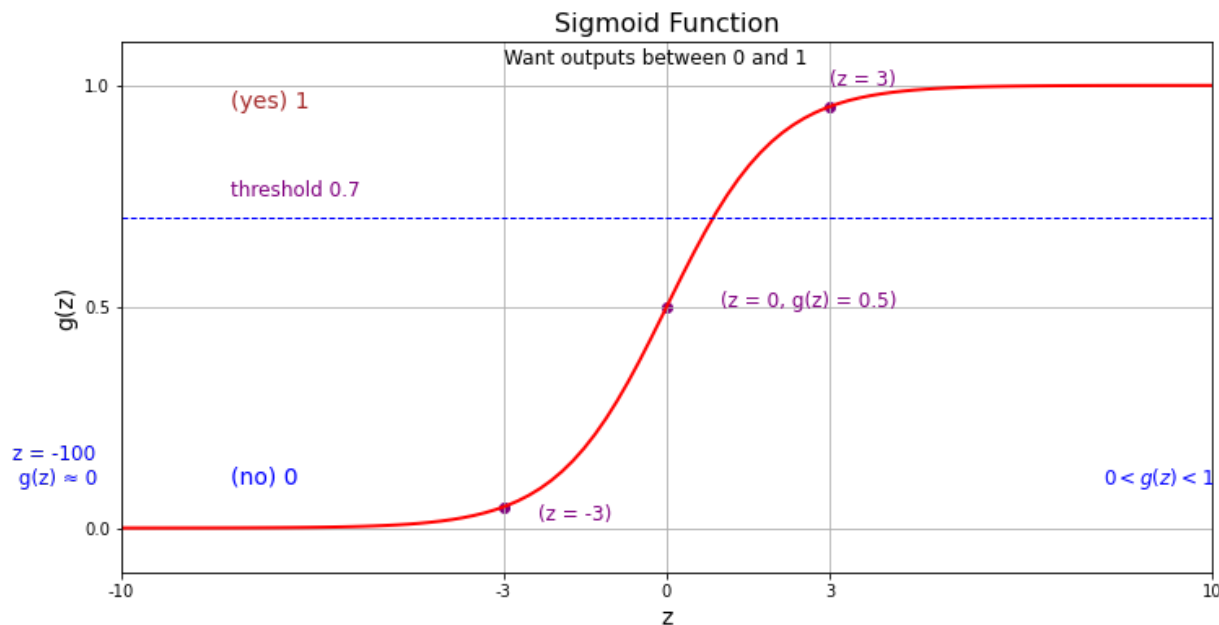
Logistic Regression Model

Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = \theta^T x \quad \text{Linear Regression}$$

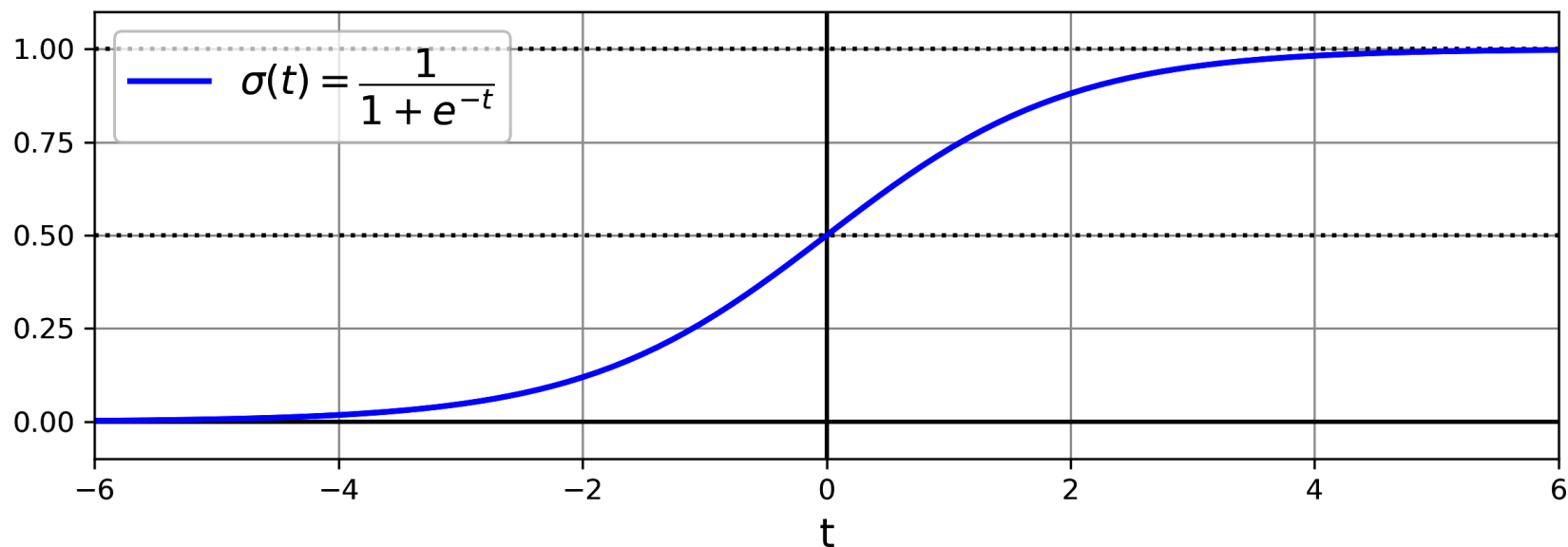
$$h_{\theta}(x) = g(\theta^T x)$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Sigmoid function
Logistic function

$$g(z) = \frac{1}{1 + e^{-z}} = \sigma(z)$$



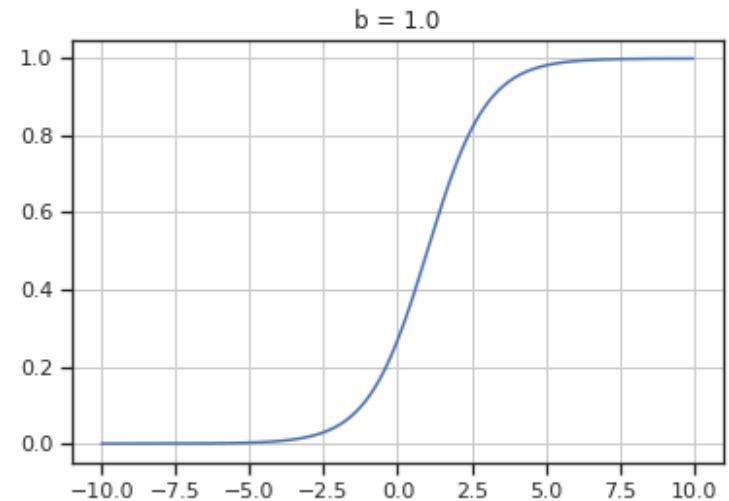
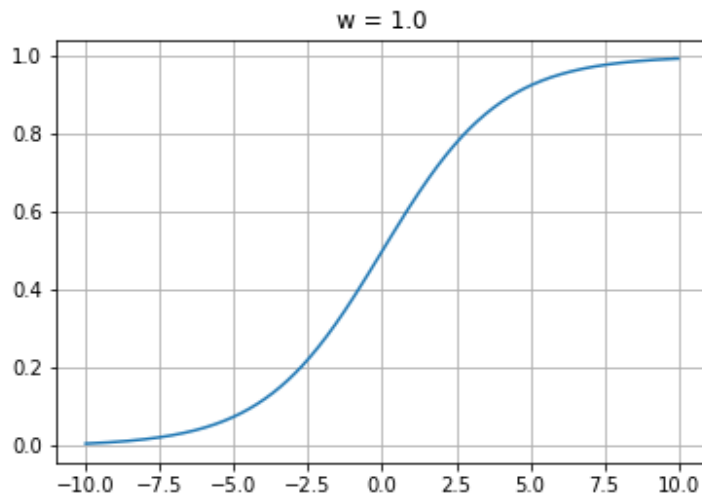
$$\sigma(t) = \frac{1}{1 + \exp(-t)}$$

The score t is often called the *logit*. The name comes from the fact that the logit function, defined as $\text{logit}(p) = \log(p / (1 - p))$, is the inverse of the logistic function. Indeed, if you compute the logit of the estimated probability p , you will find that the result is t . The logit is also called the *log-odds*, since it is the log of the ratio between the estimated probability for the positive class and the estimated probability for the negative class.

Sigmoid/Logistic Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \text{ where } z = b + \mathbf{w}^T \mathbf{x}$$

Log loss and $\mathbf{w}$ determine the steepness of the sigmoid function



$h_{\theta}(x)$ = estimated probability, p that $y = 1$ on input x

$$h_{\theta}(x) = p = \sigma(z) = \frac{1}{1 + e^{-z}},$$

where $z = \log \frac{p}{1-p} = b + \mathbf{w}^T \mathbf{x}$

z is the log-odds of the probability p .

$$p = \frac{1}{1 + e^{-z}}$$
$$1 - p = \frac{e^{-z}}{1 + e^{-z}}$$
$$\frac{p}{1-p} = \frac{\frac{1}{1+e^{-z}}}{\frac{e^{-z}}{1+e^{-z}}} = e^z$$
$$z = \log \left(\frac{p}{1-p} \right)$$

Interpretation of Hypothesis Output

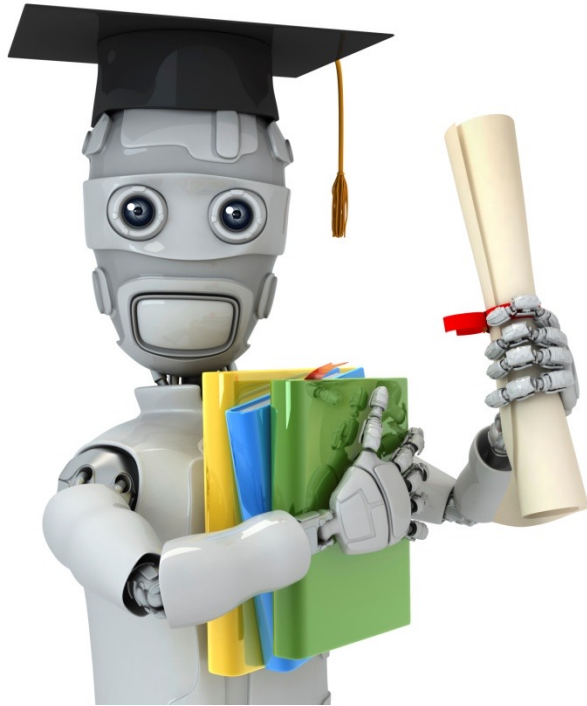
$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$
 $h_{\theta}(x) = 0.7$

Tell patient that 70% chance of tumor being malignant

“probability that $y = 1$, given x ,
parameterized by θ ”

$$P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1$$
$$P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta)$$



Machine Learning

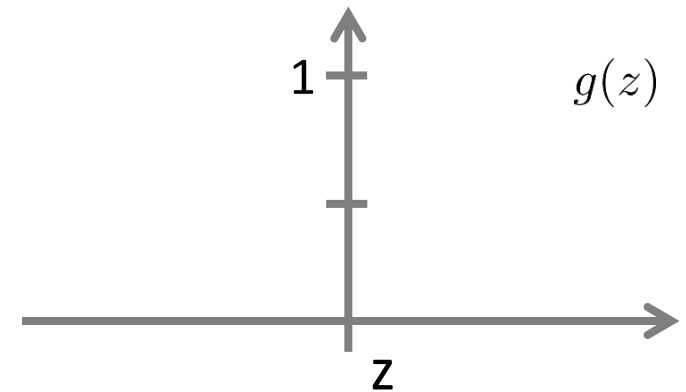
Logistic Regression

Decision boundary

Logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$

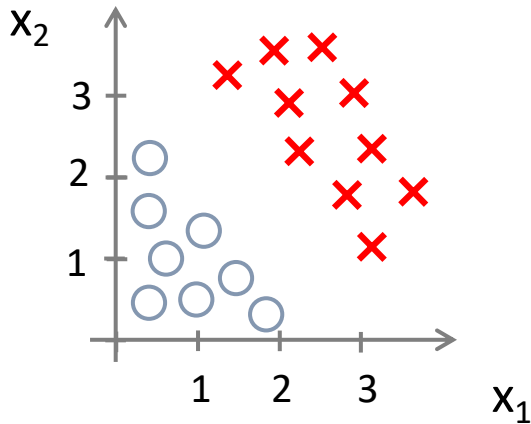


Suppose predict “ $y = 1$ ” if $h_{\theta}(x) \geq 0.5$

predict “ $y = 0$ ” if $h_{\theta}(x) < 0.5$

The logistic regression model is given by: $f_{\vec{w},b}(\vec{x}) = g(\vec{w} \cdot \vec{x} + b) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$

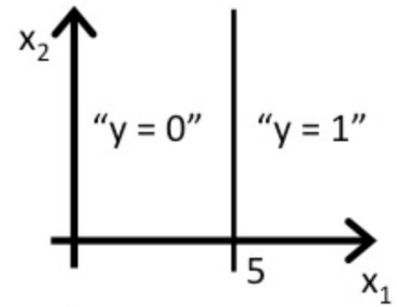
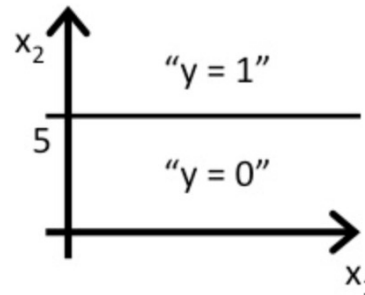
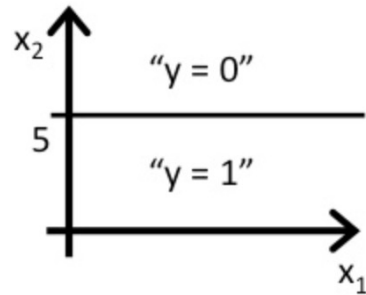
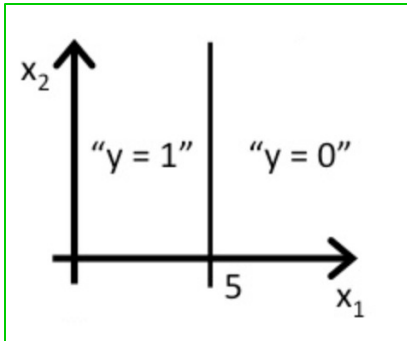
Decision Boundary



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Predict “ $y = 1$ ” if $-3 + x_1 + x_2 \geq 0$

Consider logistic regression with two features x_1 and x_2 . Suppose $\theta_0 = 5$, $\theta_1 = -1$, $\theta_2 = 0$, so that $h_\theta(x) = g(5 - x_1)$. Which of these shows the decision boundary of $h_\theta(x)$?



$$\theta = \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}$$

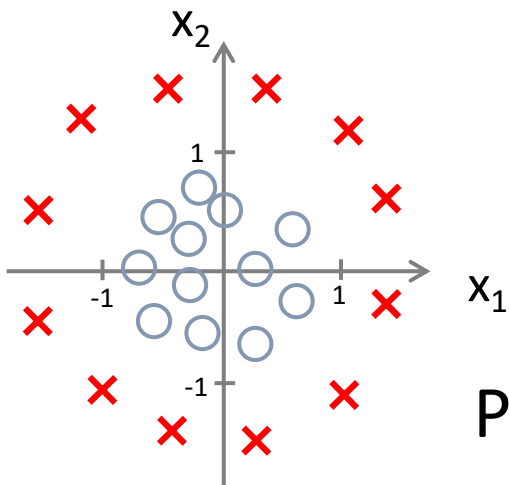
$$y = 1 \text{ if } 5 + (-1)x_1 + 0x_2 \geq 0$$

$$5 - x_1 \geq 0$$

$$-x_1 \geq -5$$

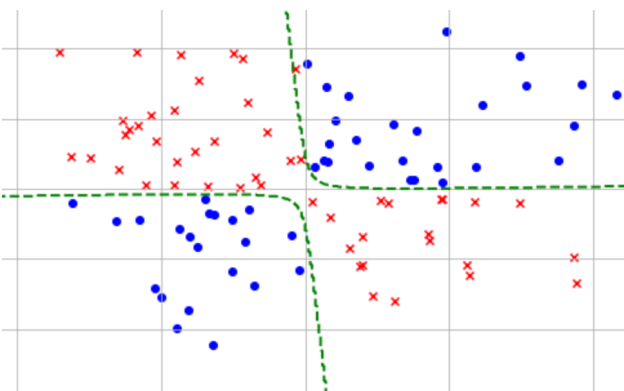
$$x_1 \leq 5$$

Non-linear decision boundaries

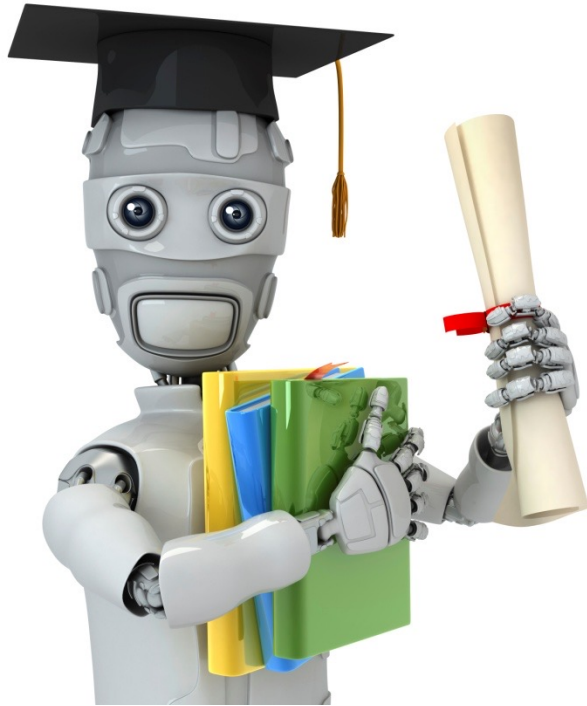


$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Predict “ $y = 1$ ” if $-1 + x_1^2 + x_2^2 \geq 0$



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \dots)$$



Machine Learning

Logistic Regression

Cost function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

m examples $x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

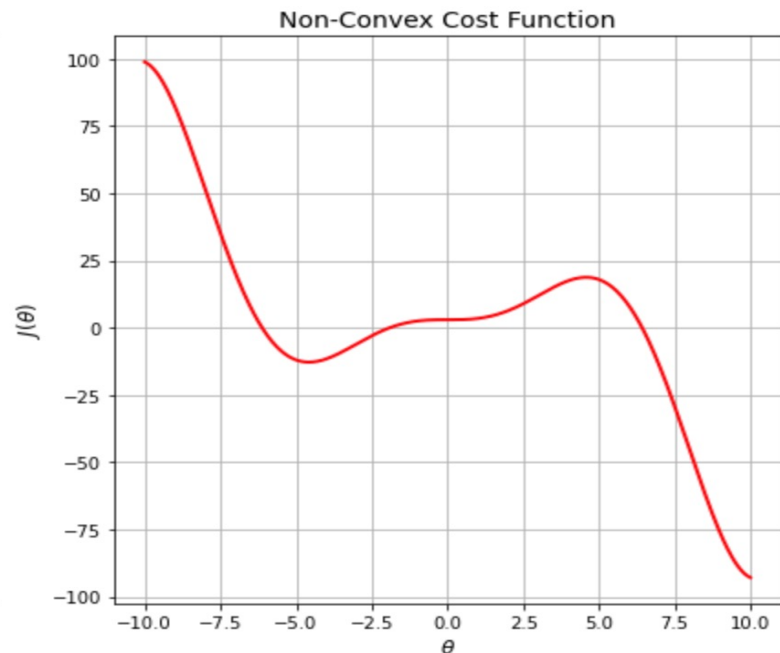
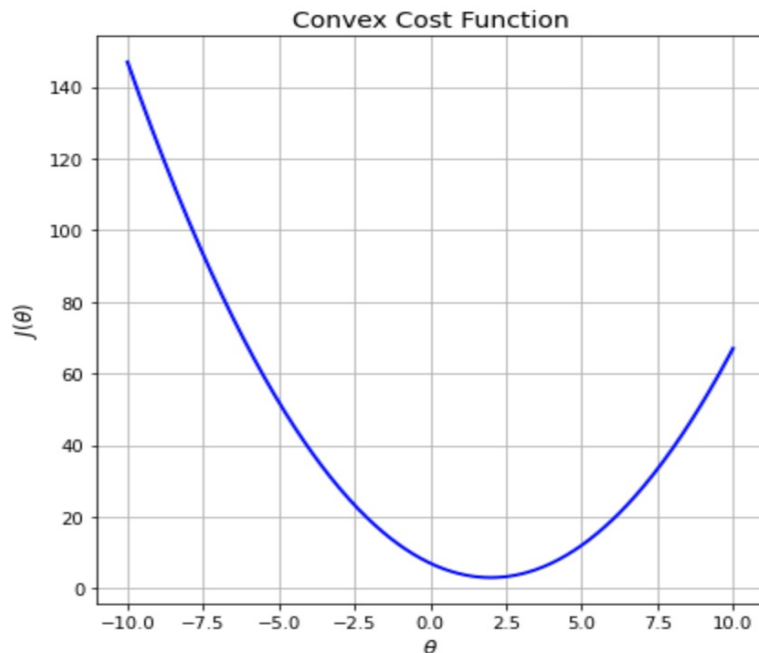
How to choose parameters θ ?

Cost function

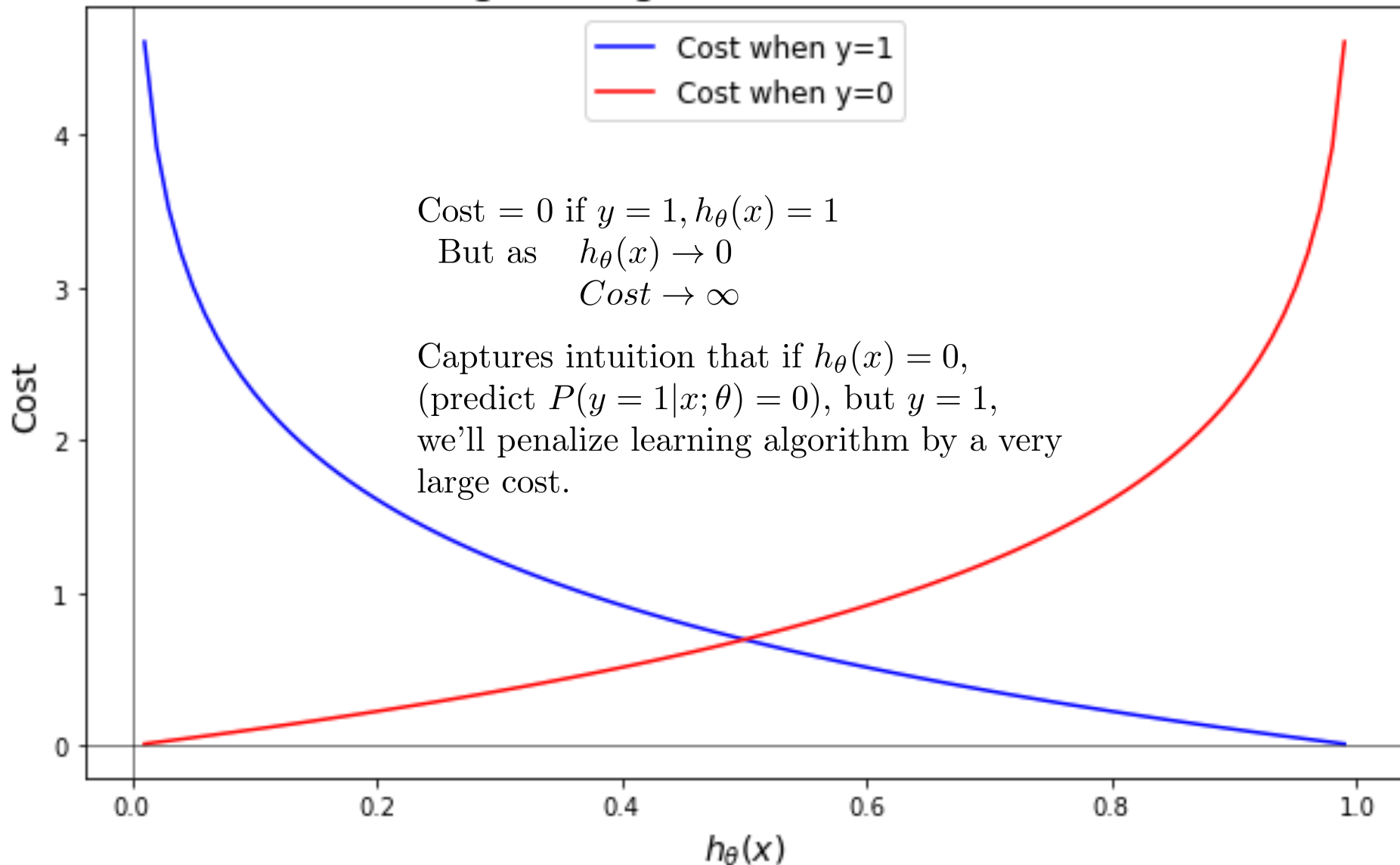
Linear regression:
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Good News: the cost function used in logistic regression is **convex**



Logistic Regression Cost Function



Graph for $\log(x)$

